



UNIVERSITI TEKNIKAL MALAYSIA MELAKA

FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGY

PROJEK SARJANA MUDA 1: PROPOSAL FORM

A TITLE OF PROPOSED PROJECT | *TAJUK PROJEK YANG DICADANGKAN*

TASMIQ AI : AI-BASED QURAN RECITATION ASSESSMENT AND MONITORING SYSTEM FOR TAHFIZ STUDENTS

B DETAILS OF STUDENT | *BUTIRAN PELAJAR*

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C PROJECT INFORMATION | *MAKLUMAT PROJEK*

(i) Executive Summary of Project Proposal [Maximum 300 words]

The proposed project, **TasmiqAI**, aims to develop an AI-based Quran recitation assessment and monitoring system to assist tahfiz students in memorizing and revising the Quran. Currently, tahfiz students depend heavily on teachers for tasmi' sessions, and students often struggle to detect tajwid and pronunciation errors during self-revision. This results in inefficient learning and limited monitoring capability.

TasmiqAI integrates artificial intelligence (AI) speech recognition, rule-based expert systems, and a web-based monitoring platform into a unified system. The AI model will be developed using **Python in the Anaconda Jupyter environment**, while **Gradio** will be used to build the AI interface and testing environment. The student mobile application will be developed using **Android Studio**, while the teacher monitoring portal will be developed using **React.js in Visual Studio Code**. The backend system and database will be implemented using **PHP and MySQL (phpMyAdmin) in XAMPP**.

Students will recite Quran verses through the mobile application, and the recorded audio will be processed by the AI system to detect tajwid and makhraj errors. The system will provide real-time feedback and accuracy scores to students. Teachers will be able to monitor student progress through a web-based dashboard that displays recitation history, performance analytics, and feedback tools.

The objectives of this project are to develop an AI-based Quran recitation assessment system, detect tajwid errors using expert system rules, provide monitoring tools for teachers, and evaluate system performance. The project scope is limited to selected surahs due to computational constraints, but the system architecture is scalable for future full Quran implementation. This project contributes to digital Islamic education by enhancing self-learning and improving recitation accuracy through AI technology.

(ii)	Detailed Proposal of the Project	
	(a) Introduction <i>(Project Background and Problem Statements)</i>	
	Recent advancements in artificial intelligence and speech recognition technologies enable the development of intelligent learning applications for Quran education. In this project, Python (Anaconda Jupyter) and Gradio are used to develop the AI recitation analysis system, while Android Studio is used to develop the student mobile application. The teacher monitoring portal is developed using React.js in Visual Studio Code, and PHP with MySQL (phpMyAdmin) in XAMPP is used for backend data management.	
	(b) Objectives of the Project	
	This project embarks on the following objectives: • To develop an AI-based Quran recitation assessment system for tahfiz students. • To implement speech recognition and expert system rules to detect tajwid and makhraj errors. • To design a mobile application that provides real-time feedback and accuracy scores to students. • To develop a web-based teacher monitoring portal to track student recitation performance and learning progress. • To evaluate the system performance using accuracy metrics and usability testing methods.	
	(c) Scope of the Project	
	1.Module to be developed (i) User Authentication and Management Module Handles student and teacher registration, login, and role-based access control. (ii) Student Mobile Application Module Allows students to record Quran recitations, view feedback, and track personal memorization progress. (iii) Speech Recognition Module Converts student recitation audio into Arabic text using pre-trained AI speech recognition models. (iv) Tajwid Expert System Module Applies rule-based reasoning to detect tajwid errors such as incorrect makhraj, missing mad extension, and incorrect verse sequences. (v) Recitation Evaluation Module Calculates accuracy scores and identifies missing or incorrect verses.	

(vi) Database Management Module

Stores user data, recitation records, scores, and teacher feedback in a MySQL database.

(vii) Teacher Monitoring Portal Module

Provides dashboards, charts, and reports for teachers to monitor student performance.

(viii) Feedback and Notification Module

Allows teachers to provide feedback and students to receive improvement suggestions.

2. Target User

- Tahfiz Students: Primary users for Quran memorization and murajaah practice.
- Tahfiz Teachers: Users who monitor student progress and provide guidance.
- Tahfiz Institutions: Organizations that require student performance monitoring and reporting tools.

(d) Expected Outcome/ Proposed Solution

The proposed solution, TasmiqAI, is an integrated system that combines artificial intelligence, mobile application technology, and a web-based monitoring platform to enhance Quran recitation learning for tahfiz students.

The system utilizes an AI-based engine consisting of Automatic Speech Recognition (ASR) and a rule-based Tajwid expert system. The ASR component processes student recitation audio and converts it into phonetic or textual representation. This output is then analyzed using predefined Tajwid rules to detect pronunciation errors, including makhraj inaccuracies and incorrect application of Tajwid rules such as Idgham, Ikhfa, and Mad.

The expected solution includes a mobile application that allows students to record Quran recitations and receive immediate feedback in the form of accuracy scores, highlighted errors, and improvement suggestions. The system also maintains a history of recitations to support continuous self-learning and revision (muraja'ah).

In addition, a web-based teacher portal will be developed to provide monitoring and evaluation capabilities. Teachers can review student recitations, access performance analytics, identify common mistakes, and provide feedback. The system supports a hybrid evaluation approach, where AI-generated scores are complemented by teacher verification to ensure accuracy and reliability.

The expected outcomes of this project include:

- Improved Quran recitation accuracy among students through AI-assisted feedback
- Reduced dependency on physical tasmi' sessions by enabling self-learning
- Enhanced monitoring capabilities for teachers through dashboards and analytics
- Identification of common Tajwid errors using data-driven insights
- A scalable system architecture that can be extended to cover the full 30 Juz Quran dataset

Overall, TasmiqAI aims to modernize traditional Quran learning by integrating AI technology with structured teacher supervision, thereby improving learning efficiency and accessibility.

D REFERENCES | RUJUKAN

State your references (Minimum 10 references)

1	Salim, M., et al. (2025). Dataset Ratio Experiment for Tajweed Law Recognition Using MFCC and CNN Features. <i>Journal of Engineering and Emerging Computing Systems</i> .
2	'Abdelfattah, O., et al. (2025). Automatic Pronunciation Error Detection and Correction of the Holy Quran's Learners Using Deep Learning. <i>arXiv preprint arXiv:2509.00094</i> .
3	Shaiakhmetov, D., et al. (2025). Evaluation of the Pronunciation of Tajweed Rules Based on Deep Neural Networks. <i>arXiv preprint arXiv:2503.23470</i> .
4	Mosul University. (2025). QDAT: A Dataset for Reciting the Quran. <i>International Journal of Advanced Science and Technology</i> .
5	Tarteel. (2024). AI-Based Quran Memorization and Recitation Application. Retrieved from https://www.tarteel.ai
6	Qara'a. (2024). AI-Based Quran Pronunciation Learning Application.
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9	Hugging Face. (2024). Quranic Audio Dataset for Speech Recognition. Retrieved from https://huggingface.co/datasets
10	Junaid Alam. (2023). Quran Speech Recognition System (QuranIVR). GitHub Repository. Retrieved from https://github.com/junaidalamgithub/QuranIVR

E DECLARATION BY STUDENT | AKUAN PELAJAR

(i)	Date: 26/3/2026	Student's Signature: 
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**E RECOMMENDED BY SUPERVISOR
PERAKUAN OLEH PENYELIA**

	Recommended
	Not Recommended
	Comments:
	Supervisor's Name:

	Signature & Stamp:
	Date: 